

Learning Human-Grounded Interaction Strategies through Action-Mediated Language Policies

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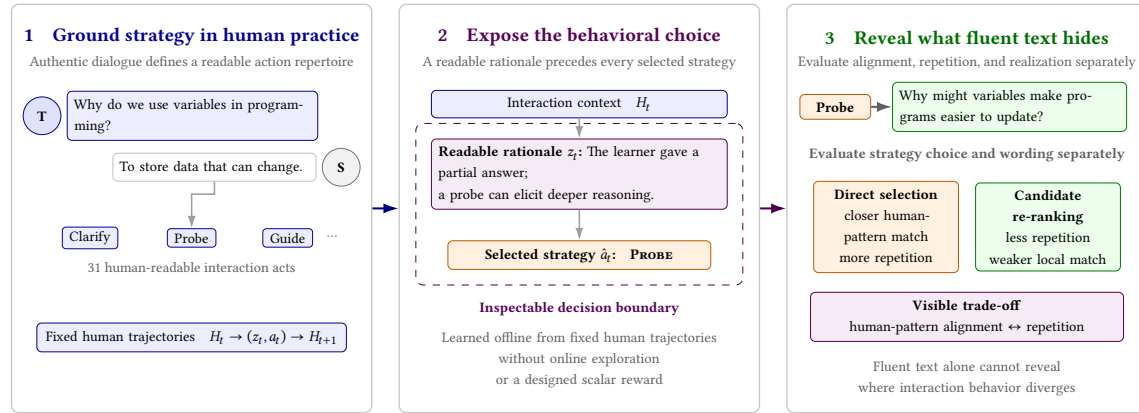


Fig. 1. **From human practice to inspectable language-mediated behavior.** Authentic dialogue grounds trajectories over readable interaction acts; LA-IQL exposes a rationale and selected strategy before language realization; and the resulting action boundary supports separate evaluation of behavioral choice and surface expression. In our classroom testbed, this boundary reveals a trade-off: direct selection more closely matches held-out human action patterns, whereas the completed candidate configuration has less repetition but lies farther from those patterns.

Human interaction is organized through strategies such as clarifying, challenging, guiding, and building on prior contributions. Language models can produce fluent turns, but the behavioral choices organizing their longer interactions remain implicit. We advance an *action-mediated* research paradigm in which authentic human interaction grounds an explicit strategy layer that can be learned, inspected, and evaluated separately from language realization. We instantiate it in Language-Action Inverse Q-Learning (LA-IQL), which learns offline without online exploration or a hand-designed scalar reward and decomposes behavior into a readable rationale, a value-based choice over human-readable interaction acts, and a separately realized utterance. Across 90 formal K–12 trajectories, the evaluated TD Direct checkpoint has 35% fewer two-step cycles and 26% fewer repeated motifs than w/o-TD Direct, while paired bigram JSD is numerically similar. A masked-label diagnostic raises action-prediction Macro-F1 from 0.0159 to 0.4044 when rationales are added, and a frozen semantic intervention flips 60% of policy actions, showing action-predictive intermediate information and sensitivity to dialogue text beyond preserved action history. Direct more closely matches held-out human action patterns than Candidate, whereas Candidate recurs less, exposing a control-dependent alignment–repetition trade-off. A metric-matched Markov control further shows that strong local alignment can be achieved without reading the dialogue. Together, these results establish action mediation as a practical way to learn human-grounded strategy policies and to separate alignment, temporal organization, context use, control allocation, and realization for analysis. The rationale and intervention results are diagnostic rather than evidence of causal faithfulness or contextual appropriateness.

CCS Concepts: • **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → *Reinforcement learning*.

Additional Key Words and Phrases: human-grounded interaction, language-mediated agents, action-mediated policy, offline imitation learning, dialogue strategy

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1 Introduction

Human interaction is organized by more than locally plausible utterances. Across an extended exchange, people question, clarify, challenge, guide, take up prior contributions, and shift initiative in ways that create recognizable interaction strategies. Language increasingly mediates how interactive assistants, multi-agent systems, and embodied agents coordinate with humans and one another [Park et al. 2023; Xi et al. 2025]. For such agents, fluent wording is important, but it does not by itself establish that the interaction is behaviorally varied, responsive, or grounded in patterns of human practice.

Large language models (LLMs) commonly generate dialogues from topics, roles, or task metadata, including through role-based multi-agent simulation [Achiam et al. 2023; Kasneci et al. 2023; Park et al. 2023; Xi et al. 2025; Zhang et al. 2025]. Their behavioral choices, however, remain implicit inside open-ended text. A plausible next sentence may clarify, challenge, guide, repeat a familiar pattern, or cede initiative, and locally fluent turns can still accumulate into repetitive interaction [Li et al. 2016a; Zhao et al. 2017]. Behavioral cloning and supervised fine-tuning provide direct ways to imitate observed examples, but optimize local action or token likelihood without explicitly modeling longer interaction dynamics. Online reinforcement learning can optimize sequential behavior, but exploration with people or high-stakes simulators may be costly or unsafe.

This gap motivates our central question: *how can language agents learn human-grounded interaction strategies from authentic dialogue while keeping those strategies explicit enough to evaluate and control?* We use *human-grounded* in a deliberately bounded sense: the action repertoire, training trajectories, and comparison patterns come from observed human interaction. The term does not imply that generated dialogue is globally human-like, natural, or effective for people.

Our central contribution is an action-mediated research paradigm in which a language policy first chooses a discrete, human-readable interaction act and then realizes that choice in language. This separates the source of a strategy—human demonstrations—from the wording of any individual turn. It also creates a traceable boundary at which strategy learning, temporal organization, control allocation, and the eventual language realization can be evaluated as related but distinct objects. Learning from fixed trajectories avoids online exploration and a hand-designed scalar reward [Li et al. 2016b; Ouyang et al. 2022; Schulman et al. 2017].

We investigate this problem through classroom discourse, a testbed containing frequent, fine-grained interaction moves. Teacher–student exchanges are multi-role and open-ended, combining questioning, uptake, elaboration, challenge, guidance, and synthesis rather than following a fixed task script. Authentic transcripts pair this richness with an established cognitive-act annotation scheme, providing an observable operationalization of interaction strategies [Song et al. 2023]. Our aim is not to model pedagogical effectiveness or replace teachers, but to study how strategy-rich human interaction can ground a language-agent policy.

We propose **Language-Action Inverse Q-Learning (LA-IQL)**, an offline framework configured by a domain-specific action space and human demonstrations. LA-IQL learns action trajectories with rationale-conditioned offline inverse soft Q-learning over fixed transitions [Garg et al. 2021; Wulfmeier et al. 2024], selects an interaction act from the learned policy, and delegates its language realization to a role model. We evaluate whether generated trajectories

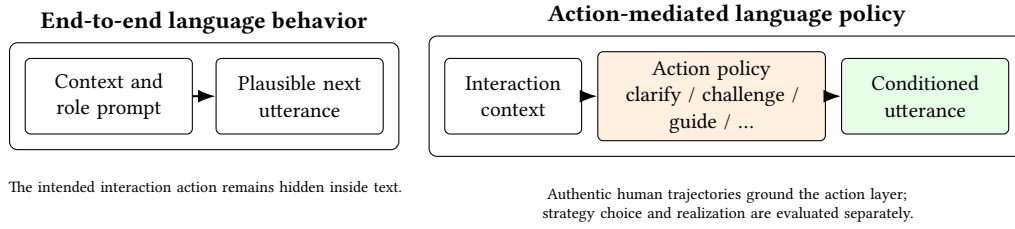


Fig. 2. Human-grounded, action-mediated interaction. LA-IQL learns a human-readable strategy layer from authentic interaction trajectories and separates strategy selection from language realization. This boundary makes behavioral patterns traceable while exposing a realization risk when the final utterance does not express the selected action.

reproduce held-out human action patterns, whether that alignment coexists with repetitive behavior, and whether rationales carry action-relevant information. We additionally define a frozen evaluation of how action-control allocation relates to observable knowledge introduction, connection, and development across the resulting long-horizon dialogues; that revised G4 analysis is not yet reported in this working draft.

Our contributions are threefold:

- We advance an action-mediated research paradigm for human-grounded interaction-strategy learning. It turns strategy into an explicit interface between authentic human trajectories and open-ended language generation, making behavioral choice, temporal organization, control allocation, and surface realization separately inspectable.
- We instantiate the paradigm in LA-IQL, a rationale-conditioned dual-head language model that combines token-level supervision, action prediction, and cross-head consistency with offline inverse Q-learning over fixed human interaction trajectories. This integration adapts temporal-difference-regularized imitation from token generation to a human-readable interaction-action policy with separate role-conditioned realization.
- We contribute a long-horizon evaluation paradigm that jointly audits human-pattern alignment, recurrence, associations between rationales and actions, semantic context sensitivity, and language realization. It reveals both a TD-associated reduction in recurrent organization and a control-dependent alignment–repetition trade-off that fluent text or aggregate similarity alone would conceal.

2 Related Work

2.1 Pedagogically Structured Dialogue and Policy Learning

End-to-end dialogue generation has progressed from sequence-to-sequence and hierarchical models to pretrained Transformers [Serban et al. 2017; Sutskever et al. 2014; Vinyals and Le 2015]. Role prompting and multi-agent simulation now support extended social, task, and classroom interactions [Park et al. 2023; Xi et al. 2025; Zhang et al. 2025], but fluent role realization does not by itself expose or control the pedagogical decision that produced an utterance. Recent educational systems therefore introduce intermediate instructional structure. BIPED derives fine-grained tutor and student dialogue acts from human tutoring data and first predicts a tutor act before generating its response [Kwon et al. 2024]. StratL steers an LLM through a prespecified multi-turn tutoring plan represented as a transition graph [Puech et al. 2025], while ScaffoldLM maintains a stepwise plan, inferred learner state, and assessment-driven memory to select tutoring actions across turns [Li et al. 2026]. Together, these systems show a shift from unconstrained response

generation toward explicit pedagogical planning and action selection. LA-IQL shares this separation of decision and realization, but learns a policy over human-annotated cognitive interaction acts from fixed classroom trajectories rather than prescribing a tutoring plan or inferring a task-specific solution path.

This separation also connects educational dialogue to dialogue-policy learning. Task-oriented dialogue is commonly formulated as sequential decision making, with abstract actions selected from dialogue states and policies optimized by reinforcement learning [Li et al. 2016b, 2019]. In educational dialogue, recent online RL aligns tutors through simulated student–tutor interaction and an explicit trade-off between pedagogical support and task success [Dinucu-Jianu et al. 2025]. Such simulation-based optimization offers controllable objectives but depends on the validity of the simulator and reward. Offline alternatives avoid new interaction: CHAI ranks language-model candidates with a learned Q-function [Verma et al. 2022], IQ-Learn recovers a policy from expert demonstrations [Garg et al. 2021], and Wulfmeier et al. [Wulfmeier et al. 2024] cast inverse soft Q-learning for language as MLE with temporal-difference regularization. LA-IQL adapts this last objective from token-level imitation to a 31-class teacher–student action policy (two roles $\times$ 15 cognitive-act symbols + END) grounded in authentic classroom trajectories [Amatari 2015; Song et al. 2023]. Its contribution is not a new IRL objective: it combines rationale-conditioned action selection, action-level TD regularization, and separate language realization without online exploration or a hand-designed scalar reward.

2.2 Proactivity, Mixed Initiative, and Control Allocation

HCI research treats conversational agency as more than response quality: an agent must decide when and how to contribute, how much control to assume, and how its behavior remains legible to people. The Inner Thoughts framework separates covert thought formation from overt participation to support proactive conversation [Liu et al. 2025]. Studies of shared autonomy find non-monotonic trade-offs: delegating communication can aid multitasking while creating discomfort, whereas shared control can impose monitoring costs [Cheng et al. 2025]. In group conversation, users also resist agents that dominate and seek controls over when, what, and where an agent contributes [Houde et al. 2025]. These works frame initiative as an allocation of control rather than a uniformly desirable increase in autonomy. Our allocation boundary is internal to the generated interaction: we compare a learned policy that selects from the full action space with proposal-constrained re-ranking in which the role model first limits the available actions. This does not test user control or establish a preferred autonomy level; it makes the policy–role-model division inspectable at the action layer.

2.3 Evaluating Pedagogical Dialogue and Simulation

Evaluation must likewise distinguish structural fidelity from pedagogical quality. MRBench provides human annotations over eight learning-science-based dimensions of tutor responses [Maurya et al. 2025]. ReWIRED annotates fine-grained teaching acts in expert explanatory dialogue and shows both that few-shot LLMs struggle to recover these acts and that functional explanation quality requires evaluation beyond static structural metrics [Anagnostopoulou et al. 2025]. EducationQ instead evaluates teaching in dynamic multi-agent scenarios and finds that general model capability does not map monotonically to teaching effectiveness [Shi et al. 2025]. Knowledge-graph representations can provide a semantic layer between dialogue utterances and grounded information items [Schneider et al. 2024], but they do not by themselves establish pedagogical value or learning.

The validity of simulated learners is a separate concern. Teachers report that prompted LLM students can be unnaturally attentive, linguistically complex, and logically inconsistent [Martynova et al. 2025]; a broader 2026 benchmark likewise finds simple prompting inadequate across linguistic, behavioral, and cognitive simulation criteria [Scarlatos

Category	Symbols	Role expansion
Prior-known knowledge	Ipk, Pk	student/teacher $\times$ symbol
Personal information	Ipi, Pi	student/teacher $\times$ symbol
Analysis	Ia, An	student/teacher $\times$ symbol
Coordination	Ic, Co	student/teacher $\times$ symbol
Speculation	Is, Sp	student/teacher $\times$ symbol
Uptake	Iu, Up	student/teacher $\times$ symbol
Agreement, querying, guide	Ag, Qu, Gu	student/teacher $\times$ symbol
END	episode termination	role-independent

Table 1. Compact definition of the 31-action interaction space.

et al. 2026]. We therefore evaluate action-pattern diversity and expert alignment, rationale–action association, and observable knowledge structures as distinct diagnostics. We do not treat generated classroom trajectories as evidence of authentic learner behavior, instructional appropriateness, or learning outcomes.

3 Method

LA-IQL treats human-grounded, language-mediated interaction as sequential decision making: at each step, the system observes dialogue history, chooses a domain-defined interaction action, and realizes that action in natural language. Following the imitation-learning perspective of offline inverse reinforcement learning, LA-IQL keeps the scalability of supervised fine-tuning while adding a dynamics-aware regularizer over expert trajectories. Our technical contribution is the integration of rationale-conditioned policy states, action-level TD-regularized imitation, and direct policy decoding with separate language realization rather than a new IRL objective. We additionally examine proposal-constrained candidate re-ranking as an inference-time adjustment.

3.1 Classroom Interaction as an Observable Testbed

We use a finite-context Markov decision process (MDP) approximation for offline policy learning from expert classroom demonstrations. Let H_t denote the full interaction history before step t . The policy observation $s_t = f_{\text{short}}(H_t)$ consists of the previous two dialogue turns and two action labels; it is an approximate Markov state that may omit longer-range instructional context. The action a_t is the next interaction act to be generated. It is drawn from a 31-action discrete space: $\{\text{student, teacher}\} \times 15$ cognitive-act symbols (e.g., basic-knowledge questioning/responding, analysis and elaboration, summarization, transfer and innovation, response construction, agreement/challenge/guidance), plus a terminal symbol END. A selected utterance u_t and its action advance the history as $H_{t+1} = H_t \oplus (a_t, u_t)$, after which $s_{t+1} = f_{\text{short}}(H_{t+1})$; END marks episode termination. Thus the deployed transition is jointly induced by the action policy and the language realization model rather than by a_t alone. Given expert demonstrations $\mathcal{D}_E = \{(s_t, a_t, s_{t+1})\}$, we learn an action policy whose generated trajectories are evaluated for diversity and similarity to expert action patterns; language realization is evaluated separately. This action-level formulation is the first LA-IQL design choice: instead of applying inverse RL directly to open-ended tokens, we optimize over human-readable teacher–student cognitive acts and leave surface realization to a separate role model.

We evaluate interaction-pattern diversity (G1), expert-pattern alignment (G2), and rationale–action consistency (G3). The revised G4 targets observable knowledge-construction patterns in generated dialogue, with its frozen system-level analysis pending in this working draft.

3.2 Data Construction and Rationale Annotation

Each expert classroom trajectory is converted into fixed-window transitions. For step t , s_t contains the two most recent utterances and the two most recent action labels, a_t is the annotated next interaction act, and s_{t+1} is obtained after appending the next utterance together with a_t . We use the deepseek-chat API (temperature 0.7) to rewrite rationale targets for 9,168 transitions from the local state and expert next action. The artifact records the model alias and call date but not an immutable provider snapshot. The trained model does not receive the expert action as an inference-time input. It instead learns the factorization

$$p_\theta(z_t, a_t | s_t) = p_\theta(z_t | s_t) \pi_\theta(a_t | s_t, z_t). \quad (1)$$

Rationale targets that copy an action symbol or canonical label are filtered before training to reduce trivial label leakage. Automated generation checks cover length, uniqueness, and retry status; they do not constitute a semantic-quality audit. We use a fixed supervision schema:

<Rationale> explanation text </Rationale> <Action> role-symbol label </Action>.

This construction turns each expert transition into paired supervision for rationale generation and action-level policy learning without a manually specified scalar reward. Because the annotation model sees the expert action whereas the deployed rationale generator does not, training mixes teacher-forced and model-generated rationales to reduce this distribution shift.

3.3 From MLE to Expert Distribution Matching

LA-IQL augments supervised fine-tuning of a rationale-conditioned dual-head model with offline IQLearn. Whereas MLE predicts the observed action locally, the IQLearn term couples the policy to successive state values without online interaction or a learned simulator.

3.4 TD-Regularized Offline IQLearn Objective

We do not introduce a new inverse reinforcement learning objective. Instead, we adopt offline inverse soft Q-learning for language imitation [Wulfmeier et al. 2024]. Grounded in IQ-Learn [Garg et al. 2021], it recasts inverse RL as maximum-likelihood training with a temporal-difference regularizer. Our contribution is its adaptation from token-level generation to a 31-class cognitive-act policy whose Q-values are produced by a rationale-conditioned policy head. We define the augmented policy state as $x_t = (s_t, z_t)$. For teacher-forced examples, $x_t^* = (s_t, z_t^*)$ and $x_{t+1}^* = (s_{t+1}, z_{t+1}^*)$ use the corresponding annotated rationale targets; model-generated-rationale examples replace these targets with samples from $p_\theta(z | s)$. At inference, the language head generates z_t from s_t before action scoring. The objective contains four terms. First, $\mathcal{L}_{\text{lm}}$ is the language-modeling loss over the LLM-annotated rationale and action prefix. Second, the policy head is supervised with action-level MLE:

$$\mathcal{L}_{\text{policy}} = -\log \pi_\theta(a_t | x_t). \quad (2)$$

Third, the LM head induces a soft action distribution $p_{\text{LM}}(\cdot | x_t)$ by scoring the fixed action-label templates after the rationale. We align the policy head with this soft distribution using a KL consistency loss:

$$\mathcal{L}_{\text{consistency}} = D_{\text{KL}}(p_{\text{LM}}(\cdot | x_t) \parallel \pi_\theta(\cdot | x_t)). \quad (3)$$

Finally, following the formulation of Wulfmeier et al., the policy head produces $Q_\theta(x_t, a)$ for every $a \in \mathcal{A}$. The corresponding soft value and policy are

$$V_\theta(x_t) = \log \sum_{a \in \mathcal{A}} \exp Q_\theta(x_t, a), \quad (4)$$

$$\log \pi_\theta(a | x_t) = Q_\theta(x_t, a) - V_\theta(x_t).$$

For an expert transition (x_t, a_t, x_{t+1}) , define the action-level temporal-difference residual:

$$\delta_t := V_\theta(x_t) + \log \pi_\theta(a_t | x_t) - \gamma V_\theta(x_{t+1}). \quad (5)$$

We set $V_\theta(x_{t+1}) = 0$ when $a_t = \text{END}$. Since $V_\theta + \log \pi_\theta = Q_\theta$, the corresponding implicit reward is $\delta_t = Q_\theta(x_t, a_t) - \gamma V_\theta(x_{t+1})$. Together, $-\log \pi_\theta(a_t | x_t) + \lambda_4 \delta_t^2$ is the action-level counterpart of the MLE-plus-TD objective derived by Wulfmeier et al. These terms are combined as

$$\mathcal{L} = \mathcal{L}_{\text{lm}} + \lambda_2 \mathcal{L}_{\text{policy}} + \lambda_3 \mathcal{L}_{\text{consistency}} + \lambda_4 \delta_t^2. \quad (6)$$

The coefficient λ_4 is zero in Stages 1–2. In Stage 3 it is set to 1 for Full and remains 0 for no-TD, recovering the non-TD objective when the term is disabled.

Training uses three successive two-epoch LoRA stages. Stage 1 emphasizes structural validity; Stage 2 jointly optimizes language and policy objectives; and Stage 3 continues joint optimization while increasing the proportion of self-generated rationales. The base model is Qwen/Qwen3.5-0.8B, the maximum sequence length is 512, and the physical batch size is 4 with one gradient-accumulation step (effective batch size 4). All three stages use training seed 42 and gradient clipping at 1.0. The evaluated Full and no-TD configurations are matched except for the Stage 3 TD-loss weight, which is 1 for Full and 0 for no-TD. Appendix B records the stage-specific values and the remaining unresolved training fields.

3.5 Rationale-Conditioned Policy Parameterization

The model uses an explicit rationale-conditioned dual-head design, which is the third LA-IQL design choice. The language head (causal LM) generates a `<Rationale>` span before the `<Action>` label; the policy head applies cross-attention pooling over the rationale span and outputs one value per action to parameterize $Q(x, a)$ and $\pi(a | x)$. Decisions are conditioned on the generated rationale rather than a single pooled representation.

Concretely, given an input token sequence:

- The LM head produces token-level scores for $\mathcal{L}_{\text{lm}}$ and, by scoring action-prefix templates over the 31 actions, yields a soft LM action distribution.
- The policy head pools the rationale text and outputs policy scores for $\mathcal{L}_{\text{policy}}$.
- A consistency loss $\mathcal{L}_{\text{consistency}}$ aligns the LM action-prefix distribution with the policy logits.

3.6 Direct Policy Decoding and Language Realization

At inference, LA-IQL separates short-context action selection from long-context language realization. Let $h_t = f_{\text{long}}(H_t)$ denote the role model’s longer view of the same history, distinct from the policy observation s_t . The policy model first generates $z_t \sim p_\theta(z | s_t)$ and directly selects

$$\hat{a}_t = \arg \max_{a \in \mathcal{A}} Q_\theta(s_t, z_t, a). \quad (7)$$

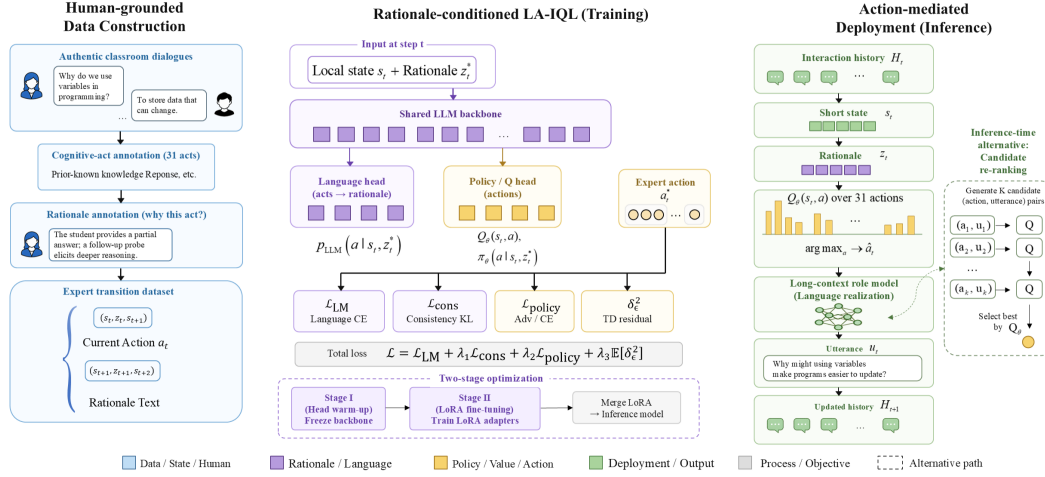


Fig. 3. **LA-IQL framework and technical flow.** Human-grounded classroom dialogue and cognitive-act annotations are converted into rationale-augmented expert transitions (left). Rationale-conditioned LA-IQL jointly trains language and policy/value components with language, consistency, policy, and temporal-difference objectives (center). At inference, the learned policy selects an interaction action and a long-context role model realizes it as an utterance; candidate re-ranking is shown as an inference-time alternative (right).

The role model then produces one realization $u_t \sim q_\phi(u | h_t, \hat{a}_t)$. Thus the default LA-IQL inference rule exposes the full action space to the learned policy while using the longer context only to realize its selected action.

The role model is recorded as deepseek-chat for both teacher and student roles. It uses a long window of 30 dialogue turns and five action steps; where sampling applies, it uses a temperature of 0.8 and top- p of 1. The artifact retains the mutable API alias rather than an immutable model snapshot. In the reported evaluation, generation stops at END or the 100-action horizon.

We additionally explored candidate re-ranking as an inference-time adjustment. The role model samples six action-utterance candidates in a fixed schema—three for the teacher and three for the student—and thereby induces $q_\phi(\mathcal{C}_t | h_t)$ over $\mathcal{C}_t = \{(a_{t,j}, u_{t,j})\}_{j=1}^6$. The selected index is

$$j^* = \arg \max_{1 \leq j \leq 6} Q_\theta(s_t, z_t, a_{t,j}),$$

$$(a_t, u_t) = (a_{t,j^*}, u_{t,j^*}). \quad (8)$$

This adjustment scores a candidate through its associated action label and does not directly score $u_{t,j}$; candidates sharing an action are therefore indistinguishable to Q_θ , and ties retain proposal order. It is proposal-constrained because an action cannot be selected unless it appears in $\mathcal{C}_t$. Unlike default direct decoding, the role model therefore controls action support while the value model only chooses among proposed actions.

Component	Recorded configuration
Policy model	Qwen/Qwen3.5-0.8B; 31-action rationale-conditioned dual head
Rationale annotation	deepseek-chat; temperature 0.7; 9,168 rewritten records
Role realization	deepseek-chat; temperature 0.8; top- p 1; 30-turn/5-action context
Policy observation	Two dialogue turns and two action labels; max input length 512; rationale budget 256 tokens
Formal split	35/5/10 lessons; subject-stage stratification; split seed 20260731
Rollout	10 test lessons; horizon 100; END disabled before step 50; margin 3.0
Randomness	One Full checkpoint and one no-TD checkpoint; generation seeds 42, 43, and 44

Table 2. Implementation summary. Values are transcribed from checkpoint, split, annotation, and formal-trajectory artifacts. Historical training fields not retained by those artifacts are listed as unresolved in Appendix B.

4 Experimental Design

Research questions. The paper is organized around four questions. This working draft reports evidence for RQ1–RQ3 and freezes the evaluation structure for RQ4: **RQ1:** How do action-mediated learned policies compare with direct language generation and statistical policies in behavioral diversity, repetition, and alignment with held-out patterns from authentic human interaction? **RQ2:** What long-horizon behavioral differences are observed between the evaluated fixed checkpoints with and without temporal-difference regularization, and between direct full-action-space selection and proposal-constrained candidate re-ranking under the completed configurations? **RQ3:** Do generated rationales contain action-relevant information beyond the dialogue state and explicit label leakage? **RQ4:** What observable patterns of knowledge introduction, connection, and development arise under different allocations of action control between the learned policy and the role model across long-horizon simulated classroom dialogues?

4.1 Data and Splits

We use 50 lessons from the K–12 classroom corpus described in Section 3.1. Each dialogue turn is associated with a speaker role and one cognitive-act symbol, from which we construct the 31 interaction acts and the transition tuples used by LA-IQL. We use a frozen, stratified lesson-level split of 35 training, five validation, and 10 test lessons rather than splitting by dialogue turn, so that windows from the same classroom session cannot occur in both partitions. The split is stratified by subject and school stage; lesson identifiers are disjoint across all three partitions, and two lessons used in preliminary smoke tests are excluded from the test partition. The split manifest, random seed, data hash, and split-level action frequencies are recorded before formal evaluation. The same test lessons were inspected in an earlier horizon-20 run; we disclose this prior access and do not alter test membership for the horizon-100 baseline evaluation reported below.

4.2 Baselines and Ablations

We compare LA-IQL with five completed controls: direct role-conditioned generation, a constrained teacher prompt, expert-frequency sampling, a first-order Markov policy, and uniform action sampling. The latter three isolate distributional and sequential statistics and are diagnostic controls rather than learned, state-conditioned policies. In particular, the Markov control estimates $p(a_t | a_{t-1})$ from the frozen training split. It is deliberately well matched to bigram- and transition-based G2 measures, but it does not observe dialogue content, the current instructional state, or a rationale, and it does not produce an utterance whose realization can be inspected. Its role is therefore to test how much local human-pattern alignment can be recovered from aggregate action transitions alone. LA-IQL uses the direct decoding

rule in Equation 7. We also report *LA-IQL + candidate re-ranking*, an inference-time adjustment that substitutes the proposal-constrained rule in Equation 8 without retraining the policy. This comparison was elevated after inspection of the current held-out results and is therefore descriptive rather than a preregistered model-selection claim. Both configurations use the same checkpoint, lessons, horizon, role-model temperature, and generation seeds. However, the completed Direct trajectories use rationale temperature 0.0 whereas Candidate uses 0.6. We therefore treat them as two system configurations and do not attribute differences solely to decoding mode. To examine the checkpoint-level difference associated with temporal-difference regularization, we add *LA-IQL w/o TD (Direct)*, trained with Stage 3 TD weight 0 rather than 1. It uses Direct decoding, rationale temperature 0.0, the same held-out lessons, role model, context windows, generation seeds, horizon, and stopping rule as the main LA-IQL system. The recovered training fields otherwise match (Appendix B), but each condition contains one trained checkpoint; we therefore treat this as a fixed-checkpoint contrast rather than a training-seed-general causal estimate.

4.3 Evaluation Metrics

For interaction-pattern diversity (G1), the main analysis reports action Distinct-2, two-step cycle rate, empirical action entropy, and expert-pattern coverage, capturing local variety, repetitive degeneration, distributional diversity, and expert-supported breadth, respectively. For expert-pattern alignment (G2), we use Jensen–Shannon divergence between generated and expert action patterns, transition-matrix distance, and expert support for generated action n -grams. These are behavioral-distribution measures, not tests of whether an action is contextually appropriate for the dialogue state. Their overlap with the Markov control’s inductive bias makes that control a demanding local-pattern reference rather than evidence of state-sensitive interaction competence. For rationale–action consistency (G3), we mask deterministic label strings and train an independent 31-way action predictor on disjoint lessons using the state, the rationale, or both. Our principal automatic measure is incremental simulatability:

$$\Delta_{\text{sim}} = F_1^{\text{macro}}(s_t, z_t \rightarrow a_t) - F_1^{\text{macro}}(s_t \rightarrow a_t), \quad (9)$$

where the target is the evaluated system’s selected action. We also report masked rationale-only Macro-F1, majority, leakage, and shuffled-rationale controls. Predictability alone does not establish causal faithfulness [Hase et al. 2020; Jacovi and Goldberg 2020]; we therefore describe G3 as a simulatability diagnostic rather than an explanation-quality measure. For secondary language-realization diagnostics, we report redundancy and rule-defined severe violations. A planned automatic action–utterance classifier did not complete its training protocol, so we exclude all results derived from that classifier. These remaining checks do not measure contextual or classroom appropriateness; open-ended text overlap is secondary because multiple responses can be valid. For the revised G4, the frozen protocol uses system–generation-seed–lesson as the statistical unit. Its primary evidence is an evidence-grounded knowledge-graph extraction followed by deterministic calculation of three prespecified endpoints: unique entities per 100 non-END turns, mean relational degree, and the proportion of final relations first introduced in the latter half of a trajectory. Rule-based diagnostics provide auditable anchors, and blinded repeated LLM judgments are auxiliary; they are not combined into a single classroom-quality score. This protocol is being evaluated separately, and no G4 system-level inference is reported in this working draft. Appendix D gives the complete metric and intervention protocol.

4.4 Evaluation Protocol

Action-pattern Jensen–Shannon divergence is the primary G2 automatic endpoint. Current trajectory results cover generation seeds 42–44 and 10 frozen test lessons per seed at horizon 100. These are three stochastic rollouts from

Method	Dist.-2 $\uparrow$	Cycle-2 $\downarrow$	H_{norm} $\uparrow$	Cov.@20 $\uparrow$
Direct Role LLM	0.4364 $\pm$ 0.0348	0.0286 $\pm$ 0.0088	0.6722 $\pm$ 0.0102	0.5500 $\pm$ 0.0408
Constrained Prompt	0.5279 $\pm$ 0.0117	0.1520 $\pm$ 0.0147	0.6818 $\pm$ 0.0015	0.5333 $\pm$ 0.0236
Uniform Random	0.9593 $\pm$ 0.0048	0.0000 $\pm$ 0.0000	0.9287 $\pm$ 0.0066	0.3500 $\pm$ 0.0000
Expert Frequency	0.6951 $\pm$ 0.0277	0.0067 $\pm$ 0.0033	0.7144 $\pm$ 0.0111	0.9500 $\pm$ 0.0000
First-order Markov	0.5144 $\pm$ 0.0072	0.0819 $\pm$ 0.0140	0.7072 $\pm$ 0.0012	1.0000 $\pm$ 0.0000
LA-IQL w/o TD (Direct)	0.4118 $\pm$ 0.0236	0.2607 $\pm$ 0.0193	0.6412 $\pm$ 0.0120	1.0000 $\pm$ 0.0000
LA-IQL	0.4370 $\pm$ 0.0101	0.1708 $\pm$ 0.0126	0.6670 $\pm$ 0.0084	1.0000 $\pm$ 0.0000
LA-IQL + candidate re-ranking	0.4478 $\pm$ 0.0205	0.0488 $\pm$ 0.0013	0.6664 $\pm$ 0.0131	0.7000 $\pm$ 0.0408

Table 3. Core G1 results (mean $\pm$ population SD over three generation seeds from each fixed checkpoint). Bold values mark the best learned-system result in each column; diagnostic baselines may be diverse without expert alignment.

Method	Paired bigram JS $\downarrow$	Pooled bigram JS $\downarrow$	Transition JS $\downarrow$	Bigram support $\uparrow$	Trigram support $\uparrow$
Direct Role LLM	0.1030 $\pm$ 0.0013	0.3423 $\pm$ 0.0184	0.1708 $\pm$ 0.0049	0.7532 $\pm$ 0.0167	0.4250 $\pm$ 0.0267
Constrained Prompt	0.1096 $\pm$ 0.0016	0.3795 $\pm$ 0.0072	0.1880 $\pm$ 0.0069	0.9218 $\pm$ 0.0092	0.4731 $\pm$ 0.0144
Uniform Random	0.1235 $\pm$ 0.0012	0.4661 $\pm$ 0.0042	0.2293 $\pm$ 0.0028	0.4105 $\pm$ 0.0249	0.0431 $\pm$ 0.0047
Expert Frequency	0.1036 $\pm$ 0.0006	0.2868 $\pm$ 0.0036	0.1995 $\pm$ 0.0036	0.8125 $\pm$ 0.0161	0.3316 $\pm$ 0.0032
First-order Markov	0.0791 $\pm$ 0.0009	0.0982 $\pm$ 0.0038	0.1249 $\pm$ 0.0017	0.9935 $\pm$ 0.0021	0.8557 $\pm$ 0.0169
LA-IQL w/o TD (Direct)	0.0835 $\pm$ 0.0022	0.1530 $\pm$ 0.0177	0.1308 $\pm$ 0.0041	0.9833 $\pm$ 0.0005	0.8773 $\pm$ 0.0125
LA-IQL	0.0827 $\pm$ 0.0031	0.1637 $\pm$ 0.0085	0.1299 $\pm$ 0.0050	0.9906 $\pm$ 0.0024	0.8772 $\pm$ 0.0064
LA-IQL + candidate re-ranking	0.1059 $\pm$ 0.0025	0.3116 $\pm$ 0.0132	0.1648 $\pm$ 0.0035	0.9522 $\pm$ 0.0045	0.4874 $\pm$ 0.0063

Table 4. G2 expert-pattern alignment (mean $\pm$ population SD over three generation seeds from each fixed checkpoint). Bold values mark learned-system results that are numerically best or effectively tied at the shown precision. The first-order Markov control directly models the local transitions measured by several columns but does not condition on dialogue content.

each fixed checkpoint, not independently trained models. We report mean $\pm$ population standard deviation across generation-seed aggregates. For learned-system comparisons, we additionally pair the 30 matching generation-seed-lesson trajectories, cluster resampling by the 10 lessons, run 10,000 bootstrap resamples, and report percentile 95% intervals. The reported TD p -values are unadjusted within an exploratory matched-checkpoint family; the Direct-Candidate table applies Benjamini-Hochberg correction over its 10 retained comparisons. The TD contrast holds Direct decoding and rationale temperature fixed. By contrast, Direct and Candidate differ in rationale temperature and were examined after observing held-out results, so that comparison remains a descriptive system-level contrast rather than a confirmatory causal decoding effect.

Table 3 reports the core G1 metrics.

Table 4 reports G2 on the same runs. Paired JS uses the frozen 31² bigram vocabulary and Jeffreys smoothing ($\alpha = 0.5$); transition JS weights rows by expert frequency, and support uses training-split n -grams.

4.5 Experimental Results

Data and implementation. The corpus contains 50 lessons, 9,419 turns, 47 topics, and 9,369 transitions. We use the frozen lesson-level split described in Section 4: 35 training, five validation, and 10 test lessons. Each system has three generation runs (seeds 42–44) from the same checkpoint, with up to 100 generated actions for each test lesson. All formal trajectories suppress END before action 50. The frequency and Markov controls use only the frozen training split.

Goal	Paired metric (TD – w/o TD)	Difference	95% CI	p
G1	Distinct-2 $\uparrow$	+0.0252	[−0.0282, 0.0688]	0.3186
G1	Cycle-2 $\downarrow$	−0.0900	[−0.1400, −0.0438]	0.0002
G1	Motif rate $\downarrow$	−0.1422	[−0.2013, −0.0889]	0.0002
G1	Normalized entropy $\uparrow$	+0.0258	[−0.0156, 0.0659]	0.2204
G2	Paired bigram JSD $\downarrow$	−0.0007	[−0.0063, 0.0043]	0.8281
G2	Transition JSD $\downarrow$	−0.0009	[−0.0091, 0.0055]	0.8581
G2	Bigram support $\uparrow$	+0.0073	[−0.00001, 0.0140]	0.0510
Secondary	Local trigram redundancy $\downarrow$	−0.00001	[−0.0042, 0.0041]	—

Table 5. Fixed-checkpoint temporal-difference contrast under Direct decoding. Values compare LA-IQL with and without the TD term over the same 30 generation-seed-lesson trajectories, clustered by 10 held-out lessons, with 10,000 bootstrap resamples. Both systems use rationale temperature 0.0 and otherwise identical generation settings. Positive differences mean the TD model is numerically larger; metric arrows indicate the desirable direction.

Main Direct and candidate re-ranking use the `sft_stage3/final` checkpoint; the w/o-TD fixed-checkpoint contrast uses `sft_stage3_woTD/final`. Direct action selection is the default inference rule, whereas candidate re-ranking is a proposal-constrained inference-time configuration. Main Direct and w/o-TD Direct both use rationale temperature 0.0 and otherwise identical generation settings. Candidate uses temperature 0.6, so its comparison with Direct is not a single-factor decoding ablation. Five of 30 w/o-TD trajectories terminate after 57–95 actions; all 30 main Direct trajectories reach 100 actions. This stopping behavior is an observed policy outcome under the shared rule, not a post-hoc truncation.

The evaluated TD checkpoint shows less repetitive long-horizon organization. Table 5 reports the matched Direct-decoding checkpoint contrast. The w/o-TD checkpoint has more two-step cycles (0.2607 vs. 0.1708; TD–w/o-TD difference −0.0900, 95% CI [−0.1400, −0.0438], $p = .0002$) and greater motif recurrence (0.5379 vs. 0.3957; difference −0.1422, 95% CI [−0.2013, −0.0889], $p = .0002$). These are 53% and 36% relative differences, respectively. Distinct-2 and normalized entropy are numerically higher with TD but do not differ reliably. Both systems cover all top-20 expert patterns, indicating that the reduction in recurrence does not coincide with abandoning those common patterns.

Local expert-distribution alignment changes little. Paired bigram JSD is 0.0827 with TD and 0.0835 without TD (difference −0.0007, 95% CI [−0.0063, 0.0043], $p = .8281$); transition JSD is likewise not reliably different. Bigram support shows a borderline descriptive advantage for TD (0.9906 vs. 0.9833, $p = .0510$). For these fixed checkpoints, the observed TD-associated difference is concentrated in recurrent sequence structure rather than local imitation fidelity. Independent training repetitions are needed to generalize this difference to the training objective.

Local transition statistics form a strong but limited reference. Tables 3 and 4 report the formal trajectory measurements. The first-order Markov control obtains the lowest paired bigram JSD (0.0791), pooled bigram JSD (0.0982), and transition-row JSD (0.1249), compared with 0.0827, 0.1637, and 0.1299 for Direct LA-IQL. It also reaches 0.9935 bigram support, while Direct LA-IQL is higher on trigram support (0.8772 vs. 0.8557).

This result follows from a substantive match between model and measurement: the control estimates $p(a_t \mid a_{t-1})$ from training trajectories, while bigram JSD and transition-row JSD measure the corresponding local action structure. High support and coverage can likewise arise from conservatively resampling training-supported transitions. These are legitimate strengths on the stated G2 endpoints, not an error to discount. However, the control never observes the dialogue text or instructional state and cannot expose whether a locally frequent transition is appropriate in context. We

Goal	Paired metric (Direct – Candidate)	Difference	95% CI	BH q
G1	Distinct-2 $\uparrow$	−0.0108	[−0.0377, 0.0165]	0.5483
G1	Cycle-2 $\downarrow$	+0.1220	[0.0986, 0.1447]	< .0001
G1	Motif rate $\downarrow$	+0.2543	[0.2100, 0.2963]	< .0001
G2	Paired bigram JSD $\downarrow$	−0.0232	[−0.0305, −0.0152]	< .0001
G2	Bigram support $\uparrow$	+0.0384	[0.0283, 0.0492]	< .0001
G2	Trigram support $\uparrow$	+0.3898	[0.3551, 0.4238]	< .0001

Table 6. Paired Direct–Candidate comparison over 30 matched generation–seed–lesson units (10,000 bootstrap resamples; percentile intervals; BH correction over 10 retained comparisons). Positive values mean Direct is numerically larger; metric arrows indicate the desirable direction. The configurations also differ in rationale temperature, so this is a system-level comparison rather than a single-factor decoding ablation.

therefore interpret it as evidence that local human-pattern reproduction is achievable with a simple sequential statistic, and not as evidence that the Markov policy is a more capable interactive agent. This distinction motivates reporting G2 alongside state-conditioned decision traces, repetition diagnostics, and action–utterance realization checks.

A frozen intervention provides exploratory evidence of text sensitivity. We evaluated the main Full checkpoint on five pre-existing, evenly spaced states from each of the 10 held-out lessons ($N = 50$). Within each lesson, a cyclic donor replaced utterance semantics while preserving the speaker sequence, number of utterances, and action history. Relative to the original states, the probability assigned to the expert action decreased by 0.0767 (lesson-cluster bootstrap 95% CI [0.0153, 0.1288]), 60% of argmax actions flipped (CI [0.48, 0.72]), and mean policy JSD was 0.2957 (CI [0.2476, 0.3431]). The 31-class Macro-F1 decrease was 0.0201, with an interval crossing zero ([−0.0185, 0.0649]). This intervention indicates that the fixed policy distribution responds to dialogue text beyond the preserved action history. Its small sample, deterministic donor construction, and lack of human judgments mean it does not establish that the response is contextually appropriate. Appendix C records the protocol.

Within the learned architecture, alignment comes with repetitive local structure. Direct LA-IQL is closer to the held-out expert distribution than Candidate: paired bigram JSD is 0.0827 versus 0.1059, and training-expert bigram support is 0.9906 versus 0.9522. Across 30 matched generation–seed–lesson units, the Direct–Candidate difference in bigram JSD is −0.0232, 95% CI [−0.0305, −0.0152], BH-adjusted $q < .0001$.

This alignment does not imply uniformly better interaction dynamics. Direct has substantially more two-step cycles (0.1708 vs. 0.0488; paired difference +0.1220, 95% CI [0.0986, 0.1447], $q < .0001$) and repeated motifs (0.3957 vs. 0.1413; difference +0.2543, 95% CI [0.2100, 0.2963], $q < .0001$). Distinct-2 has no reliable paired difference (−0.0108, 95% CI [−0.0377, 0.0165], $q = .5483$). The configurations therefore differ more clearly in action–sequence organization than in local combinatorial diversity.

A rule-selected trace makes the recurrence difference inspectable. Before reading trajectory text, we selected the matched lesson–seed unit with the largest Direct–Candidate Cycle-2 difference, breaking any tie by motif-rate difference and then lexical order. This rule selects the Grade-9 chemistry lesson on atomic structure at generation seed 43 (Cycle-2: 0.309 vs. 0.052; motif rate: 0.55 vs. 0.17). Table 7 shows the earliest four-turn Direct cycle and the same turn positions in the matched Candidate trajectory. Rationales and utterances are abridged translations from Chinese; action symbols follow Table 1.

Configuration and local context	Four-turn decision trace: action; rationale → utterance
Direct	
Student had just restated the teacher's explanation of nuclear force.	t13 Teacher-Ag: confirm understanding and extend the issue → "Exactly. Does having more protons mean stronger nuclear force and a more stable nucleus?" t14 Student-Pk: answer the hypothesis from prior knowledge → "I think so; more protons should mean more nuclear force." t15 Teacher-Ag: affirm the analogy, then introduce a competing force → "But electrostatic repulsion also increases; which force wins?" t16 Student-Pk: respond to that comparison → "Repulsion also grows, so the nucleus may not necessarily become more stable."
Candidate	
Student had just estimated electron mass and located most atomic mass in the nucleus.	t13 Teacher-Gu: confirm the estimate and consolidate mass-distribution understanding → "Draw an atomic model and mark each part's position and approximate mass share." t14 Student-An: construct and explain the requested model → "A small central nucleus contains protons and neutrons; light electrons move outside it." t15 Teacher-Iu: refine the diagram's component placement and mass analogy → "What force keeps the electrons and nucleus in a stable relation?" t16 Student-Pk: supply the requested relation → "The attraction between negative electrons and the positive nucleus."

Table 7. Rule-selected inspectable case from formal trajectories (lesson 4873, generation seed 43, turns 13–16). The Direct window instantiates an Ag-Pk-Ag-Pk two-step cycle, whereas Candidate traverses Gu-An-Iu-Pk. The entries preserve the recorded action, rationale intent, and utterance content while abbreviating the original text for space.

The action trace exposes recurring decision types that would be difficult to audit from surface text alone. At the same time, the Direct utterances advance from a claim about nuclear force to the countervailing role of electrostatic repulsion. The Candidate trace also exposes a local rationale–realization shift at t15: the recorded rationale focuses on refining placement and mass, while the utterance asks about electrostatic stability. The example therefore illustrates, but does not by itself validate, the aggregate recurrence contrast or realization quality: a repeated action motif is not equivalent to verbatim repetition or instructional failure. Determining whether either interaction is more coherent or useful still requires human judgment.

Rationales carry action-predictive information. Under the same G3-v2 dual TF-IDF evaluator, masked rationale-only Macro-F1 is 0.3986 for Direct and 0.1013 for Candidate, compared with state-only scores of 0.0159 and 0.0123. Generation-seed-level incremental simulatability Δ_{sim} is 0.3885 for Direct and 0.0863 for Candidate. The w/o-TD Direct checkpoint obtains rationale-only Macro-F1 0.3682, combined Macro-F1 0.3629, $\Delta_{\text{sim}} = 0.3372$, and accuracy 0.6867, descriptively below the corresponding TD Direct values of 0.3986, 0.4044, 0.3885, and 0.7097. For the original Direct and Candidate configurations, the observed generation-seed-level delta exceeds all 20 within-lesson/role shuffle controls, and all 30 generation-seed-lesson deltas are positive. These results show that masked rationales contain information about the selected action. They do not show that rationales causally determine the policy decision or constitute faithful explanations; we also do not attach an inferential TD-effect claim to the descriptive G3 difference.

Language realization remains a separate risk. The planned G4-v1 action–utterance classifier did not complete its training protocol. We therefore exclude its preliminary agreement rates and all classifier-based system comparisons from the reported evidence. The remaining rule-based checks show no exact repetitions or rule-defined severe violations for Direct, Candidate, or w/o-TD Direct, while local trigram redundancy does not differ reliably between Direct and Candidate or between the two Direct checkpoints. These narrow checks do not measure realization quality or classroom appropriateness.

RQ4 knowledge-construction results are pending. The frozen G4 structure will compare the prespecified knowledge-graph endpoints across matched system-generation-seed-lesson units and will report lesson-cluster uncertainty. Extractor and judge quality gates, system-level estimates, and the planned growth curve have not yet been completed. We therefore leave the numerical table and figure unpopulated and make no claim about which configuration produces richer, better connected, or more educationally valuable knowledge construction.

Appendix A gives the full system-level comparison, G3-v2 table, and judgment status.

5 Discussion

5.1 Human Grounding Becomes an Inspectable Design Resource

Our first result is methodological: action mediation makes human grounding an operational and inspectable design resource. The action vocabulary, training demonstrations, and held-out comparison patterns all come from authentic classroom interaction, connecting policy learning, generation, and evaluation through the same human-readable strategy layer. This is more than a structured prompt for next-utterance generation: it provides a stable interface at which designers can inspect what strategy was selected, how strategies organize over time, and how each choice is realized in language.

The metric-matched Markov control strengthens the case for this paradigm. Its strong local alignment shows that transition similarity can be reproduced without reading the dialogue; the semantic intervention, in contrast, shows that the learned policy distribution responds to dialogue text beyond the preserved action history. Human-pattern similarity is therefore useful when it is situated inside a broader, inspectable evaluation rather than treated as a complete measure of context-sensitive interaction. Human grounding still does not guarantee globally human-like or contextually appropriate behavior, but the action boundary makes the relevant successes and failures empirically separable.

5.2 TD-Associated Differences Reveal Longer-Horizon Organization

The matched Direct-decoding checkpoint contrast localizes the clearest observed TD-associated difference. The w/o-TD checkpoint shows no reliably detected difference in local expert-pattern alignment but has 53% more two-step cycles and 36% more repeated motifs. This pattern is consistent with the TD term affecting recurrent sequence organization rather than merely expanding the action vocabulary or improving one-step imitation, but the single checkpoint per condition prevents a training-objective-level causal conclusion. The w/o-TD policy also terminates five of 30 trajectories before the 100-action horizon, whereas the TD policy terminates none, providing further descriptive evidence of different rollout behavior. Because each condition contains one trained checkpoint, the bootstrap captures held-out lesson and generation variation but not training-seed uncertainty.

This result demonstrates the analytical leverage of the action-mediated formulation. It localizes the observed TD-associated difference in temporal organization—35% fewer two-step cycles and 26% fewer repeated motifs for the evaluated TD checkpoint—rather than in local expert-pattern alignment or action-vocabulary expansion. End-to-end text evaluation would make this distinction difficult to recover. The present metrics do not establish that participants would perceive the TD trajectories as more coherent or useful; that claim requires human evaluation of complete interactions.

5.3 Action Mediation Makes Control Allocation Observable

Separating strategy selection from language realization turns control allocation into an observable design variable. The completed Direct configuration gives the learned policy access to the full action space and has stronger alignment with

held-out human action patterns, but also more short cycles and repeated motifs. The Candidate configuration restricts action support to role-model proposals and has fewer repetitions, but its trajectories are farther from the held-out action distribution. Because the configurations also differ in rationale temperature, this is a descriptive system-level contrast rather than a causal decoding ablation. Nevertheless, the completed configurations illustrate that greater policy control over action support need not coincide with lower recurrence: strategy support, temporal organization, and wording must be considered together.

The explicit action boundary provides a concrete substrate for structural traceability. It lets system designers compare the strategy selected for a state with the utterance eventually produced, inspect recurring control patterns, and localize whether a divergence originates in policy choice or realization. This is a method-level capability, not yet a validated user-facing explanation; whether people can use the boundary effectively for oversight or intervention remains an HCI question requiring direct study.

5.4 A Research Paradigm for Strategy-Rich Language Agents

Three implications follow from the present testbed. First, action-pattern metrics should be paired with controls whose inductive biases match those metrics; otherwise, local statistical reproduction may be mistaken for context-sensitive competence. Second, evaluation should treat strategy choice and language realization as separate but coupled objects: an appropriate action label is not useful when the utterance fails to express it. Third, long-horizon repetition must be reported alongside diversity and alignment, because a policy can cover human-supported patterns while repeatedly cycling among a small subset. Together, these checks define a reusable research paradigm for language agents that learn from human interaction: ground a readable strategy layer in human practice, learn and expose the policy over that layer, and evaluate its alignment, temporal organization, context use, and realization separately. The classroom instantiation demonstrates the value of this decomposition without assuming that any single similarity score establishes human-like behavior.

6 Conclusion

We advanced an action-mediated research paradigm for learning and evaluating human-grounded interaction strategies. Its central move is to place a readable strategy layer between dialogue context and language realization, turning otherwise implicit behavioral choices into objects that can be learned, inspected, and evaluated over long horizons. LA-IQL operationalizes this paradigm through rationale-conditioned action selection, temporal-difference-regularized offline imitation, and separate role-conditioned realization.

The completed evidence demonstrates the leverage of this formulation. For the evaluated fixed checkpoints, TD Direct has 35% fewer two-step cycles and 26% fewer repeated motifs than w/o-TD Direct, localizing the observed difference in longer-horizon organization while local expert-pattern alignment remains numerically similar. Direct selection more closely matches held-out human action patterns than proposal-constrained Candidate, while Candidate recurs less, making control allocation and its alignment–repetition trade-off directly visible. Masked-label diagnostics show that generated rationales carry substantial action-predictive information beyond state-only features, and the semantic intervention shows that the policy responds to dialogue text beyond preserved action history. Finally, the metric-matched Markov control confirms the need for the broader paradigm by showing that local human-pattern alignment alone can be achieved without context use.

These results do not establish a training-seed-general TD effect, causal rationale faithfulness, contextual appropriateness, or human-facing utility; the frozen knowledge-construction analysis also remains to be completed. Nevertheless,

they establish the core contribution: action mediation provides a practical bridge from authentic human interaction to inspectable policy learning, while supplying the measurement boundary needed to distinguish alignment, temporal organization, context use, control allocation, and language realization. This paradigm opens a concrete path toward strategy-rich language agents whose behavior can be studied and shaped at a level more informative than fluent text alone.

Limitations

LA-IQL relies on a fixed interaction-act taxonomy derived from one classroom annotation scheme. The resulting 31-action space may omit pedagogically meaningful behaviors and may not transfer unchanged across subjects, grade levels, languages, cultures, or classroom formats. Although the framework is designed around domain-specific state and action spaces and is motivated by a broader interaction-learning problem, this paper evaluates only its classroom-dialogue instantiation; transfer to conversational, multi-agent, and embodied systems remains a hypothesis to be established with domain-specific demonstrations and direct empirical evaluation. Because the method learns from offline expert demonstrations, it can reproduce annotation errors and systematic biases in the source corpus and cannot identify effective strategies that are absent from that corpus. The finite-context policy observation may miss long-range instructional goals even though the role model receives a longer dialogue history.

The completed Direct and Candidate configurations differ in rationale sampling temperature in addition to their action-selection mechanism. Their comparison therefore identifies a reproducible system-level trade-off but does not isolate the causal effect of decoding mode. A matched-temperature comparison is needed before making that attribution.

Generated rationales should not be interpreted as faithful causal explanations of the model’s internal computation. They are supervised textual intermediates and can be plausible while remaining incomplete or inconsistent with the selected action. The explored proposal-constrained adjustment is limited by candidate coverage: a strong Q-function cannot select an action absent from the proposal set, and it does not distinguish utterances sharing the same action label. Direct decoding avoids this support restriction but delegates realization of its selected action to a separate role model, which may fail to express that action faithfully. Finally, automatic action-distribution metrics do not establish educational effectiveness. Claims about learning outcomes or teacher-training utility require evaluation with educators and learners in realistic settings. The pending G4 protocol likewise measures knowledge structures expressed in generated text, not changes in student cognition, factual correctness, instructional quality, or learning outcomes. Its extractor and LLM-judge outputs require the frozen schema, evidence, corruption, and stability gates before they can support even system-level automatic comparisons. Moreover, the current G2 endpoints emphasize local action n -grams and transition distributions. A policy such as the first-order Markov control can score highly by conservatively reusing training-supported transitions without responding to the semantic content of the current classroom state. Our 50-state intervention provides exploratory evidence that the Full policy distribution changes with utterance semantics, but its deterministic donor and small sample do not establish contextual decision quality; broader interventions and direct state-matched judgments remain necessary.

Ethical Considerations

Classroom dialogue can contain personal information and involve minors, making privacy, consent, and data governance central to this work. Any use of classroom transcripts should follow the source dataset’s consent conditions,

license, and access restrictions, and should remove identifying information before model training or release. Generated dialogue must not be presented as a substitute for qualified teachers or used for high-stakes assessment without human oversight.

The system may reproduce cultural, linguistic, or pedagogical biases present in the demonstrations and in the underlying language models. In particular, a policy trained on a narrow classroom population may incorrectly treat one discourse style as universally expert. Deployment should therefore include subgroup and cross-context evaluation, educator review, clear disclosure that outputs are machine-generated, and mechanisms for users to contest or override generated recommendations. LLM-generated rationale annotations also require quality auditing because fluent explanations can conceal labeling errors or unsupported assumptions.

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Split	Lessons	Turns	Topics	Transitions
Train	35	6,352	33	6,317
Validation	5	1,039	5	1,034
Test	10	2,028	10	2,018
Total	50	9,419	47	9,369

Table 8. Corpus statistics for the frozen lesson-level split. Topic counts within rows are distinct counts and need not sum to the corpus total because one topic name occurs in more than one partition.

Variant	D2 $\uparrow$	Cycle-2 $\downarrow$	Motif $\downarrow$	JSD $\downarrow$
LA-IQL w/o TD (Direct)	0.4118	0.2607	0.5379	0.0835
LA-IQL Direct	0.4370	0.1708	0.3957	0.0827
Candidate re-ranking	0.4478	0.0488	0.1413	0.1059

Table 9. Training and inference comparisons on the same held-out lessons. JSD is the G2 paired lesson-level bigram divergence.

Method	State F1	Masked R-only F1	Combined F1	Δ_{sim}	Accuracy
LA-IQL w/o TD (Direct)	0.0257	0.3682	0.3629	0.3372	0.6867
LA-IQL Direct	0.0159	0.3986	0.4044	0.3885	0.7097
Candidate re-ranking	0.0123	0.1013	0.0986	0.0863	0.2393

Table 10. G3-v2 rationale–action simulatability. Values are means of the three generation-seed-level scores. The target is each system’s selected action; the result is an association diagnostic, not a causal-faithfulness measure.

A Additional Experimental Results

Training and inference configurations. LA-IQL directly selects the highest-valued action and then obtains one role-model realization. Candidate re-ranking instead generates six action–utterance proposals and constrains the policy to actions represented in that set. This changes the allocation of control between the policy and role model. It does not retrain the policy. The completed configurations also use different rationale temperatures (0.0 vs. 0.6), so the comparison below is a descriptive system-level contrast rather than an isolated decoding ablation. The w/o-TD Direct condition instead removes only the temporal-difference term from training and retains the main Direct generation configuration, including rationale temperature 0.0. Five of its 30 trajectories terminate after the minimum allowed horizon, compared with none for main Direct.

Figures 4 and 5 provide complementary visual summaries of these comparisons. The first emphasizes paired uncertainty, whereas the second preserves the reported metric scales and generation-seed-level variation.

G3-v2 rationale–action simulatability. The independent predictor masks action symbols, canonical names, abbreviations, and deterministic label strings. State and rationale use separate character 2–5-gram TF–IDF representations and validation-selected linear classifiers. All three systems use the same frozen evaluator protocol.

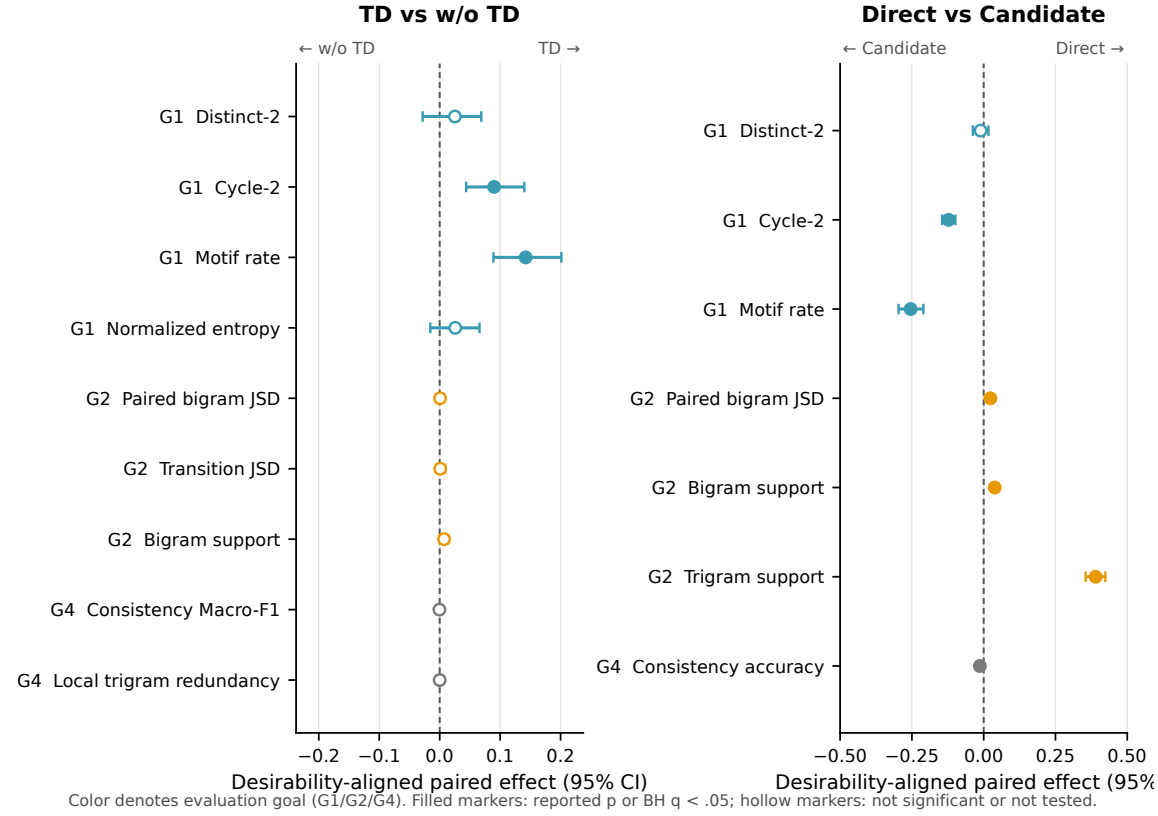


Fig. 4. Desirability-aligned paired effects for the fixed-checkpoint TD contrast and Direct–Candidate system contrast. For metrics marked ↓ in the result tables, the reported difference and interval are multiplied by -1 so that positive values consistently favor the first named system (TD or Direct). Horizontal bars are percentile 95% intervals from 10,000 paired bootstrap resamples clustered by the 10 held-out lessons. Filled markers denote a reported $p < .05$ or BH-adjusted $q < .05$; hollow markers denote a non-significant or untested comparison. The Direct–Candidate panel is a descriptive system-level comparison because rationale temperature also differs between configurations.

Judgment status. Blinded items, randomized manifests, and the rating rubric have been generated, but no completed human judgment scores are present. The incomplete G4-v1 classifier results are excluded; only the independent rule-based diagnostics are retained. We make no claim about factuality, safety, pedagogical appropriateness, experienced naturalness, or learning outcomes from the zero severe-rule violation rate.

Complete G1 diversity and recurrence diagnostics. Table 11 reports the full G1 suite for every completed system. Cov.@20 is the fraction of the 20 expert bigram or trigram patterns with greatest lesson support (with occurrence count as a tie-breaker) that appears in a generated run. These diagnostics must be interpreted jointly: high diversity or low recurrence alone does not imply expert-pattern alignment.

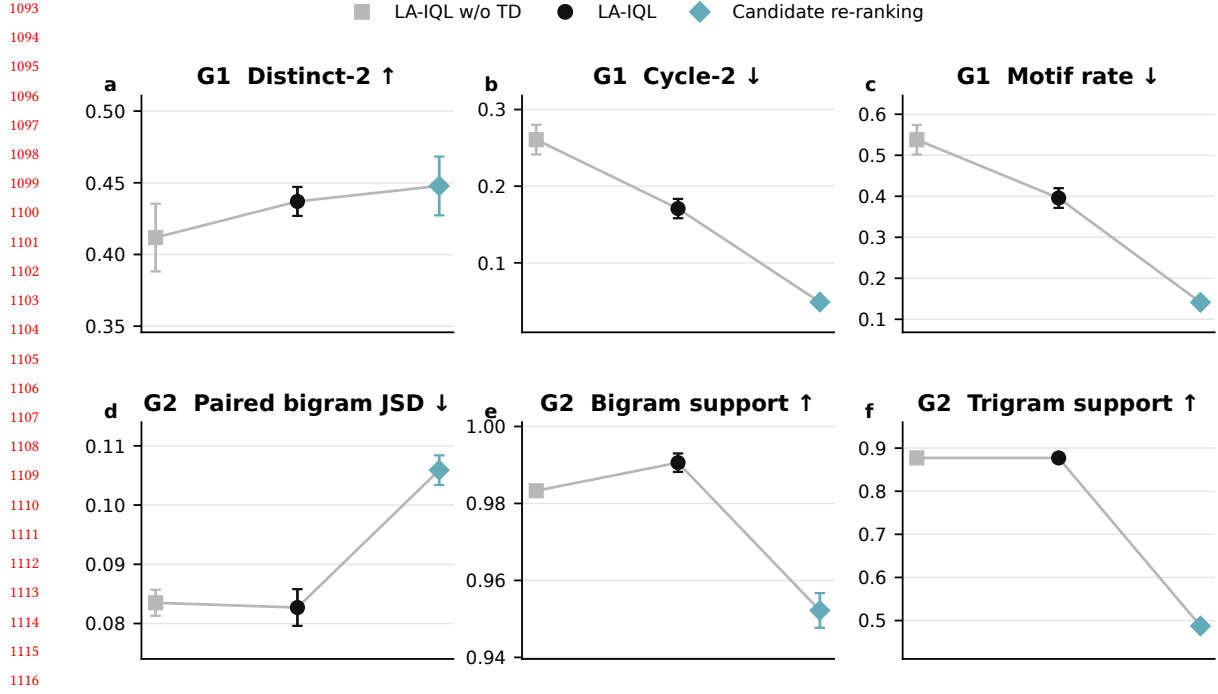


Fig. 5. Core G1 and G2 results for the three learned-system configurations. Points show means and error bars show population standard deviations across generation seeds 42–44 from each fixed checkpoint, with 10 held-out lessons per seed. Connecting lines are visual guides rather than temporal trajectories; arrows indicate the desirable metric direction.

Method	Runs	Dist.-1	Dist.-2	Dist.-3	Motif ↓	Cycle-2 ↓	Cov.@20	H_{norm}
Direct Role LLM	10×3	0.2692 ± 0.0050	0.4364 ± 0.0348	0.5913 ± 0.0427	0.4486 ± 0.0287	0.0286 ± 0.0088	0.5500 ± 0.0408	0.6722 ± 0.0102
Constrained Prompt	10×3	0.2866 ± 0.0029	0.5279 ± 0.0117	0.7264 ± 0.0144	0.3403 ± 0.0204	0.1520 ± 0.0147	0.5333 ± 0.0236	0.6818 ± 0.0015
Uniform Random	10×3	0.4149 ± 0.0198	0.9593 ± 0.0048	0.9986 ± 0.0013	0.0048 ± 0.0012	0.0000 ± 0.0000	0.3500 ± 0.0000	0.9287 ± 0.0066
Expert Frequency	10×3	0.2154 ± 0.0145	0.6951 ± 0.0277	0.9449 ± 0.0069	0.0795 ± 0.0098	0.0067 ± 0.0033	0.9500 ± 0.0000	0.7144 ± 0.0111
First-order Markov	10×3	0.2004 ± 0.0047	0.5144 ± 0.0072	0.7815 ± 0.0026	0.2136 ± 0.0417	0.0819 ± 0.0140	1.0000 ± 0.0000	0.7072 ± 0.0012
LA-IQL w/o TD (Direct)	10×3	0.1606 ± 0.0088	0.4118 ± 0.0236	0.6352 ± 0.0219	0.5379 ± 0.0361	0.2607 ± 0.0193	1.0000 ± 0.0000	0.6412 ± 0.0120
LA-IQL	10×3	0.1663 ± 0.0034	0.4370 ± 0.0101	0.6983 ± 0.0050	0.3957 ± 0.0240	0.1708 ± 0.0126	1.0000 ± 0.0000	0.6670 ± 0.0084
LA-IQL + candidate re-ranking	10×3	0.1493 ± 0.0070	0.4478 ± 0.0205	0.7221 ± 0.0238	0.1413 ± 0.0017	0.0488 ± 0.0013	0.7000 ± 0.0408	0.6664 ± 0.0131

Table 11. Complete G1 diversity and recurrence diagnostics for all completed systems. Values are mean ± population standard deviation across generation seeds 42, 43, and 44; each seed contains 10 held-out lessons. Learned systems reuse one checkpoint per condition across those seeds. Diagnostic baselines are included to show that high diversity or low recurrence alone does not establish expert-pattern alignment.

B Implementation and Reproducibility

Recovered training configuration and unresolved fields. We transcribed inference settings from the terminal config object in the formal trajectory JSON files, data provenance from the frozen split manifest, and model identity from checkpoint metadata. These artifacts are authoritative for the reported runs. The recovered run configuration confirms three two-epoch LoRA stages, the learning rates and loss weights below, physical batch size 4, one gradient-accumulation step, maximum length 512, gradient clipping at 1.0, and training seed 42. Full and no-TD share this

Stage	Initialization	Epochs	LR	Policy	Cons.	LM-act	TD Full/no-TD	Teacher/self
1	Base model	2	10^{-4}	0.25	0	1.0	0/0	1.0/0.0
2	Stage 1 final	2	10^{-4}	1.0	0.1	0.5	0/0	0.8/0.2
3	Stage 2 final	2	5×10^{-5}	1.0	0.1	0.5	1/0	0.6/0.4

Table 12. Recovered three-stage configuration for the evaluated Full and no-TD checkpoints. “Cons.” is the cross-head consistency-loss weight and “LM-act” is the LM action-class loss weight. The two configurations differ only in the Stage 3 TD-loss weight among the recovered fields.

System	Checkpoint / objective	TD	Selection	Rat. temp.	Role proposals	Randomness
Full Direct	sft_stage3/final; four-term objective	on	full-space argmax	0.0	one realization	gen. 42–44
Full Candidate	same Full checkpoint	on	six-proposal re-rank	0.6	3/role	gen. 42–44
No-TD Direct	sft_stage3_woTD/final; TD removed	off	full-space argmax	0.0	one realization	gen. 42–44
Frozen BC	sft_bc_seed*; next-action CE	off	full-space argmax	n/a	n/a	train 42–44
LoRA BC	lora_policy_seed*; next-action CE	off	full-space argmax	n/a	n/a	train 42–44

Table 13. System provenance matrix. “gen.” denotes generation seeds applied to a fixed checkpoint; “train” denotes independently trained BC checkpoints. The BC models are completed diagnostics but are not included in the formal horizon-100 trajectory tables.

configuration except that the Stage 3 TD-loss weight is 1 for Full and 0 for no-TD. The optimizer details, learning-rate scheduler and warm-up, LoRA rank/alpha/dropout and target modules, numerical precision, exact sample and optimizer-step counts, model/tokenizer revision, checkpoint-selection rule, and historical code revision remain to be verified. We do not substitute current defaults for those fields or describe the result as replicated across training seeds.

Across Stages 1–3, the recovery-loss weights are 0.5, 0.5, and 0.75; the action-token weights are 8, 5, and 5; the structure-token weights are 10, 8, and 8; the rationale-end-token weights are 50, 30, and 30; and the action-boundary-token weights are 30, 20, and 20. The TD discount is 0.99. The rationale generator uses maximum 256 new tokens and temperature 0 during training. Configuration comments that describe micro-batch 1 with accumulation 4 are stale; the actual run parameters are physical batch 4 and accumulation 1.

Completed BC training metadata. The frozen-backbone BC checkpoints use Qwen/Qwen3.5-0.8B, 6,181 training samples, next-action cross-entropy, three epochs, batch size 4, learning rate 2×10^{-5} , and maximum length 512. The LoRA BC checkpoints use the same backbone, objective, and training set plus 999 validation samples; they use three epochs, batch size 32, learning rate 10^{-4} , maximum length 256, rank 8, bfloat16, and one gradient-accumulation step. Each family retains independent training seeds 42–44 and per-epoch metadata.

Inference configuration and quality gates. All reported Full, Candidate, and no-TD formal trajectories use a two-turn, two-action policy window; a 30-turn, five-action role window; policy maximum length 512; rationale budget 256 new tokens; horizon 100; role temperature 0.8; three candidates per role where candidate re-ranking applies; and END suppression through step 49 with logit margin 3.0. The Full and no-TD generation gates each contain 50 samples and report 100% structural validity, closed rationale/action tags, no degeneration, and no length-limit hits. These checks establish schema validity, not policy or rationale quality.

Software environment. The locked environment requires Python 3.10 or newer and resolves PyTorch 2.11.0+cu128, Transformers 5.14.1, PEFT 0.20.0, scikit-learn 1.7.2, SciPy 1.15.3, OpenAI 2.52.0, and Google GenAI 2.16.0. The dependency lockfile and CUDA 12.8 wheel index are part of the code artifact.

Endpoint	Original	Counterfactual	Difference / shift
Accuracy	0.4000	0.3000	−0.1000
Macro-F1 (31)	0.1486	0.1285	−0.0201
Expert-action probability	0.2732	0.1965	−0.0767
Prediction flip rate	—	—	0.6000
Policy JSD	—	—	0.2957

Table 14. Exploratory context intervention for the main Full checkpoint ($N = 50$ states, 10 lessons). Lesson-cluster intervals appear in the main text. The intervention tests distributional sensitivity, not contextual correctness.

C Context-Sensitivity Intervention

The frozen context-sensitivity-v1 protocol selects five pre-existing, evenly spaced held-out states per lesson from the frozen G4 candidate pools. For each state, it constructs a within-lesson cyclic semantic donor while preserving the number and sequence of speakers and the complete action history. Rationales are greedily decoded with temperature 0 and a 160-token budget; END suppression is not applied because this is a state-level test. Intervals use 10,000 percentile bootstrap resamples clustered by lesson with seed 20260822.

D Evaluation Details

The complete G1 suite additionally includes Distinct-1/3, repeated local motifs, and cross-trajectory variance. For G3, masking removes action symbols, canonical names, abbreviations, and deterministic label strings before prediction. We report leakage rates, per-class scores, confusion matrices, and majority and shuffled-rationale controls.

The current G3 evaluation provides association evidence through independent prediction and within-lesson/role shuffle controls. Remove, paraphrase, and role-compatible counterfactual interventions [DeYoung et al. 2020] remain future tests of causal sensitivity. Blinded action-support and classroom- appropriateness materials have been prepared, but no completed human judgment scores are reported. We therefore avoid claims about explanation faithfulness, educational effectiveness, or experienced naturalness.